UNIT IV

17. Discuss the following :

(a) Demand paging

(b) Thrashing.

Or

18. Consider a file system where a file can be deleted and its disk space reclaimed while links to that file still exists. What problems may occur if a new file is created in the same storage area or with the same absolute path name? How can these problems be avoided?

UNIT V

19. What are the various Disk space allocation methods? Explain any two in detail.

Or

20. LINUX coordinates the activities of the kernel I/O components my manipulating shared in-kernel data structures, whereas WINDOWS uses object oriented message passing between kernel I/O components. Discuss pros and cons of each approach.

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B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2019.

Fifth Semester

CSE

OPERATING SYSTEMS

(2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. Why the operating system is viewed as the resource allocator?

2. Write the difference between job and process scheduler.

3. Define the term Thread.

4. Draw the diagrammatic view of a monitor.

5. State the conditions under which a deadlock situation may arise.

6. Why page size is always the power of 2?

7. What do you mean by best fit?

8. List down the types of accesses methods.

9. Name the FOUR registers of I/O port.

10. State the swap space management technique.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE question from each Unit.

UNIT I

11. Differentiate between Multiprocessor system and Distributed System.

Or

12. Explain the various types of system calls with an example for each.

UNIT II

13. Suppose that the following processing arrive for execution at the time indicated :

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Find the average waiting and turnaround time for these processes with :

(a) FCFS scheduling algorithm

(b) Pre-emptive SJF algorithm.

Or

14. Outline a solution using semaphores to solve Dining Philosopher problem.

UNIT III

15. Consider a system consisting of 'm' resources of same type being shared by 'n' processes. Resources can be requested and released by processes only one at a time. Show that the system is deadlock free if the following two conditions hold :

(a) The maximum need of each process is between 1 and m resources.

(b) The sum of all maximum needs is $m + n$.

Or

16. Discuss the situation under which the most frequently used page replacement algorithm generates fewer page faults than the least recently used page replacement algorithm. Also discuss under which circumstances, the opposite holds.

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B.Tech. DEGREE EXAMINATION, MAY 2019.

Fifth Semester

Computer Science Engineering

OPERATING SYSTEMS

(2013 – 14 onwards Regulations)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is Microkernel?

2. Draw the process state diagram.

3. Explain with a suitable example about critical section.

4. What are monitors?

5. Define swapping.

6. What is segmentation with paging?

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B.Tech. DEGREE EXAMINATION, MAY 2019.

Fifth Semester

Computer Science Engineering

OPERATING SYSTEMS

(2013 – 14 onwards Regulations)

Time : Three hours | Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is Microkernel?

2. Draw the process state diagram.

3. Explain with a suitable example about critical section.

4. What are monitors?

5. Define swapping.

6. What is segmentation with paging?

7. Under what circumstances do page faults occurs?

8. What is the cause of thrashing?

9. What is a Boot Control Block?

10. List out the four main object types defined by the Linux VFS.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Discuss, with examples, how the problem of maintaining coherence of cached data manifests itself in the following processing environments:

(a) Single-processor systems

(b) Multiprocessor systems

(c) Distributed systems.

Or

12. Describe the PIPEs IPC mechanism.

UNIT V

19. (a) Explain the structure of the file system in Linux and windows systems. (6)

(b) Write short notes on caching. (5)

Or

20. Explain the process management in Linux system. (11)

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B.Tech DEGREE EXAMINATION, MAY 2018.

Fifth Semester

Computer Science and Engineering

OPERATING SYSTEMS

(2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are the objectives of an operating system?

2. What are the five categories of system calls?

3. Define thread.

4. What is semaphore?

5. What are the three methods that deal with the deadlock problem?

6. Write short notes on swapping.

7. Define virtual memory.

8. What is thrashing?

9. What is internal fragmentation?

10. Define buffer in I/O system?

PART B — (5 × 11 = 55 marks)

Answer ALL question ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Explain the differences between operating system for mainframe computers and personal computers? (6)

(b) Why are distributed systems desirable? (5)

Or

12. (a) Explain the possible operations that could be applied on process in detail. (5)

(b) What is IPC? Explain (6)

UNIT II

13. (a) Explain briefly about round robin schedulin with example.

(b) What is binary semaphore? How will y implement it?

Or

14. What is critical section problem? How will yo find the solution for it? (1

UNIT III

15. Discuss briefly about Bankers algorithm wi example. (1

Or

16. What is paging? Explain briefly. (1

UNIT IV

17. What is meant demand paging? Explain how t allocation of frames can be done? (1

Or

18. (a) Explain the different file access methods file management.

(b) Explain the operations performed on directory.